

Executive Summary: Geometric Foundations for Unified Physics: An Emergent Dimensional Framework

1. Purpose & Motivation

The challenge of unifying general relativity (GR) with quantum mechanics (QM) remains one of the central open problems in theoretical physics. Incompatibility between the smooth, geometric structure of spacetime in GR and the probabilistic, discrete phenomena of QM leads to unresolved singularities—most notably in black holes and the early universe. This work aims to achieve unification through a minimal geometric completion, introducing no extra fields, particles, or ad hoc symmetries.

2. Core Concept

The framework proposes that three foundational constraints—Zero (collapse), Infinity (expansion), and Chance (probability)—emerge as structural dimensions alongside the conventional axes of space and time. Rather than being additional coordinates postulated by fiat, these emergent constraints naturally encode the domains of collapse, expansion, and quantum uncertainty directly into the metric structure of spacetime.

3. Mathematical Structure

The metric is augmented to include curvature components associated with each emergent dimension (see Appendices for explicit forms).

- Zero introduces a boundary condition for collapse, regularizing singularities.
- Infinity provides a geometric basis for cosmic expansion without recourse to dark energy.
- Chance introduces a probabilistic scaling function that geometrizes quantum fluctuations.

This extension preserves general covariance and known conservation laws. When the new parameters are taken to their classical limits (collapsed or suppressed), the model reduces cleanly to standard GR, ensuring compatibility with all established tests.

4. Physical Implications

- Resolution of Singularities: The framework provides a geometric cutoff to black hole and cosmological singularities without requiring quantization of the metric.
- Unified Quantum-Geometric Behavior: The “Chance” dimension replaces arbitrary quantization with a geometric probability field, offering a natural account of quantum phenomena.

- Geometric Cosmic Acceleration: The “Infinity” constraint explains accelerated expansion as an intrinsic property of the metric, obviating the need for a cosmological constant or dark energy.

5. Testable Predictions & Falsifiability

The framework predicts several measurable, explicitly falsifiable effects:

- Gravitational wave polarization anomalies—deviations in waveforms that differ from standard GR.
- Quantum interference patterns—geometry-driven modifications to phase correlations and vacuum fluctuation spectra.
- Vacuum energy deviations—altered predictions for observed “zero-point” energy in cosmological and laboratory contexts.

All predictions are accessible to near-term or ongoing experimental and observational programs.

6. Relation to Prior Work

Unlike string theory or extra-dimensional models, the present approach does not posit hidden axes or new gauge symmetries.

- Minimalist: All added structure is emergent from geometric necessity, not conjecture.
- Comparative: The framework differs from Kaluza-Klein, Penrose’s conformal cyclic cosmology, or loop quantum gravity, yet shares their ambition for a purely geometric unification. Key differences are discussed in the full text and appendices.

7. Conclusion & Invitation

The framework achieves clean reduction to GR in appropriate limits, introducing no violations of established physics.

It is presented as an open invitation for rigorous critique, dialogue, and technical review—particularly with regard to falsifiability and potential extensions.

Full derivations, equations, and appendix references are available on request.

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